

BIOPHARMACEUTICS (PH 411)

COURSE CONTENTS

Unit I. BCS classification of drugs.

Unit II. Biopharmaceutical evaluation of tablets, capsules, liquid orals, parenterals, dermatological.

Unit III. Bioavailability in drug development: Explanation of terms: Bioavailability (Absolute and Relative), Purpose and methods of bioavailability studies using blood level and urinary excretion data, Federal Requirements.

Unit IV. Bioequivalence, chemical and clinical and therapeutic equivalence, pharmaceutical alternative;

Unit V. Assessment methods of pharmaceutical and therapeutic equivalence; possible reasons for non-equivalence.

Unit VI. Principles of Drug Dissolution Related to Bioavailability and bioequivalence; Dissolution Rate; Elementary idea of *in vitro* and *in vivo* correlation and its significance.

Unit VII. Photostability studies, stability studies of drug products and API.

Reference Books:

"Modern Pharmaceutics", G.S.Banker and C.T.Rhodes, Marcel Dekker Inc., NY.

J.G.Wagner – Fundamentals of Clinical Pharmacokinetics, Drug Intelligence Publications, Hamilton, PA, USA.

Leon Shargel, Andrew B.C. Yu Applied Biopharmaceutics & Pharmacokinetics